Problem Set 3

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Due: November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

   I regress *voteshare* on *difflog* using the following commands in R:

```
reg_voteshare_difflog <- lm(inc.sub$voteshare ~ inc.sub$difflog, data=inc.sub)
reg_voteshare_difflog
summary(reg_voteshare_difflog)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.26832 -0.05345 -0.00377  0.04780  0.32749

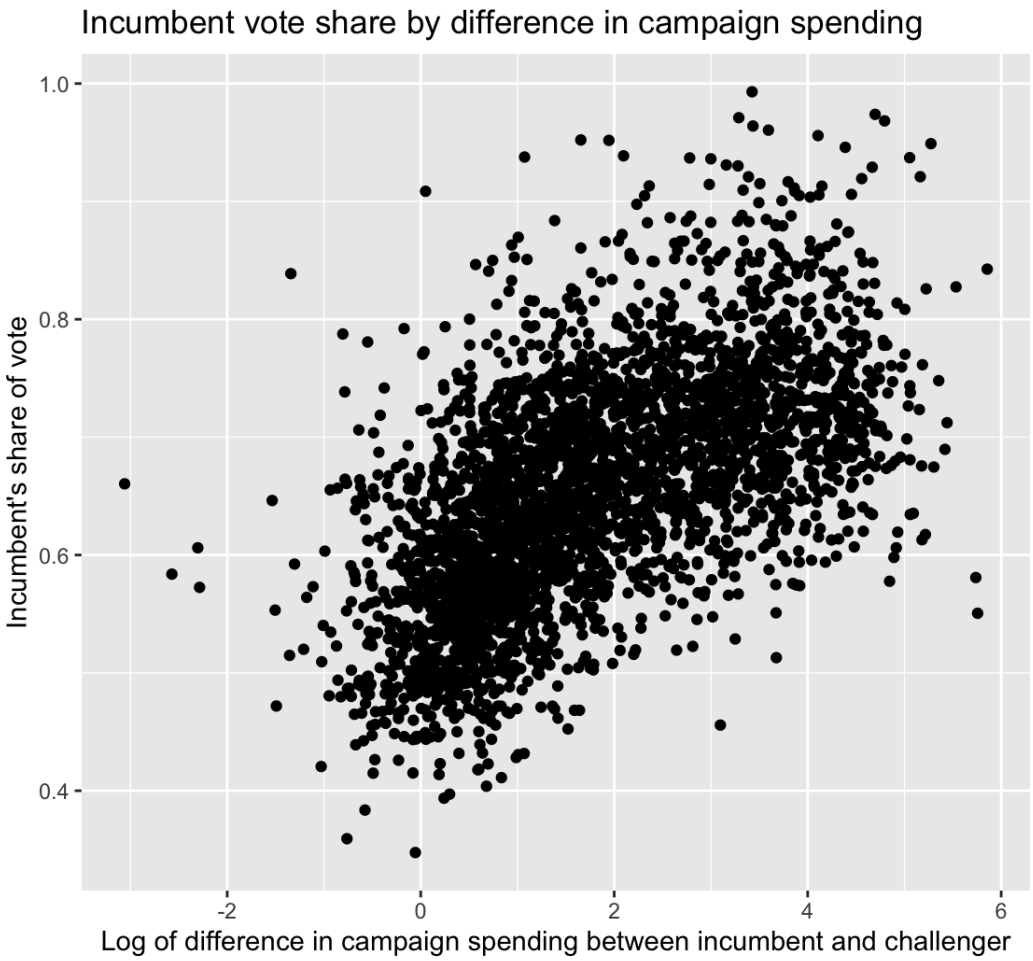
Coefficients:
Estimate Std. Error t value
(Intercept)     0.579031   0.002251  257.19
inc.sub$difflog 0.041666   0.000968   43.04
Pr(>|t|)
(Intercept)       <2e-16 ***
inc.sub$difflog   <2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic:  1853 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

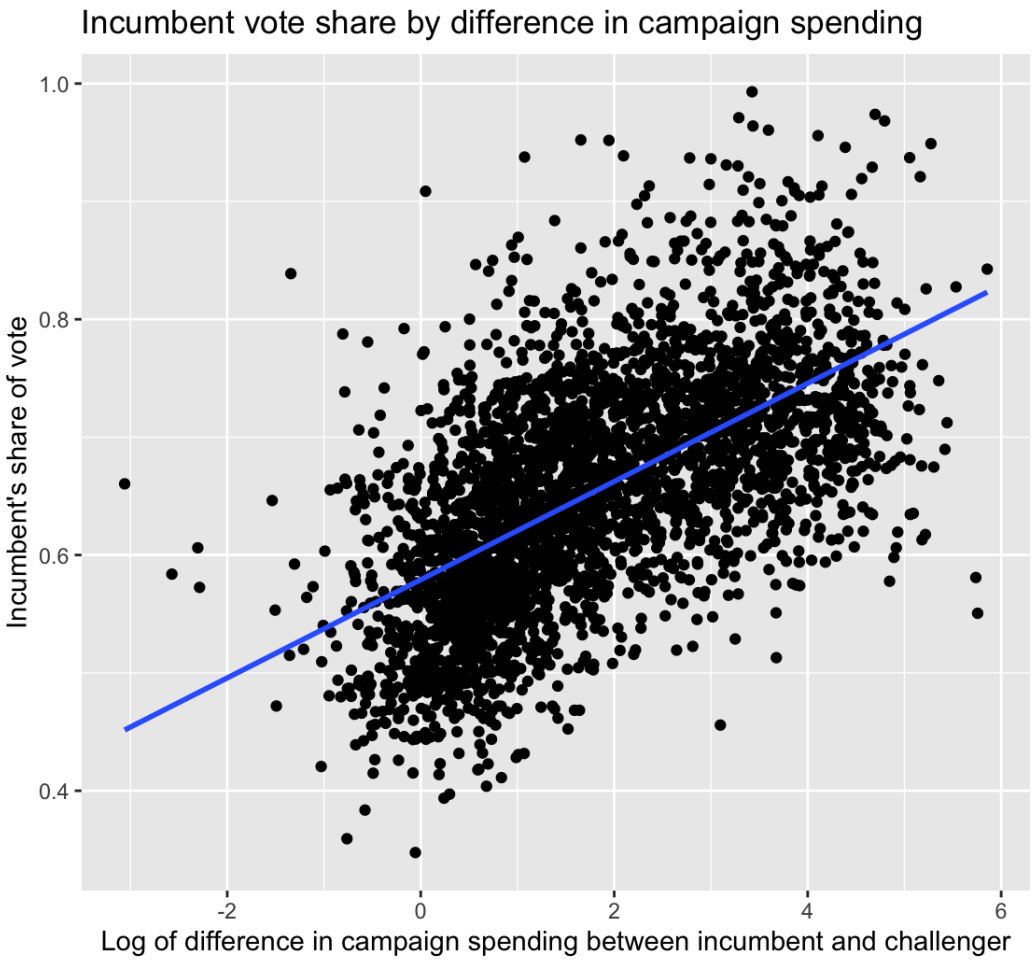
   First, I create the scatterplot using the following code:

```r
library(ggplot2)
ggplot(inc.sub, aes(x=difflog, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent vote share by difference in campaign spending",
       x = "Log of difference in campaign spending between incumbent and challenger",
       y = "Incumbent's share of vote")
```



I add the regression line using this code:

```
ggplot(inc.sub, aes(x=difflog, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent vote share by difference in campaign spending",
       x = "Log of difference in campaign spending between incumbent and
    challenger",
       y = "Incumbent's share of vote") +
  geom_smooth(
    aes(x=difflog, y=voteshare),
    method = "lm",
    formula = y~x,
    se=FALSE)
```



3. Save the residuals of the model in a separate object.

   I save the residuals of the model in a separate object using this command:

```
residuals_voteshare_difflog <- resid(reg_voteshare_difflog)
```

4. Write the prediction equation.

   I again run the "summary" command to get the regression coefficient values:

```
summary(reg_voteshare_difflog)
```

```
Residuals:
Min       1Q   Median       3Q      Max
-0.26832 -0.05345 -0.00377  0.04780  0.32749

Coefficients:
Estimate Std. Error t value
```

```
(Intercept)     0.579031   0.002251  257.19
inc.sub$difflog 0.041666   0.000968   43.04
Pr(>|t|)
(Intercept)       <2e-16 ***
inc.sub$difflog   <2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic:  1853 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) = 0.579031

$\hat{\beta}_1$ (slope) = 0.041666

Therefore:

$\hat{voteshare} = \mathbf{0.579031 + 0.041666(\textit{difflog})} + \epsilon$

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

   I regress *presvote* on *difflog* using the below code:

```
reg_presvote_difflog <- lm(inc.sub$presvote ~ inc.sub$difflog, data=inc.sub)
reg_presvote_difflog
summary(reg_presvote_difflog)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.32196 -0.07407 -0.00102  0.07151  0.42743

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.507583   0.003161  160.60   <2e-16 ***
inc.sub$difflog 0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

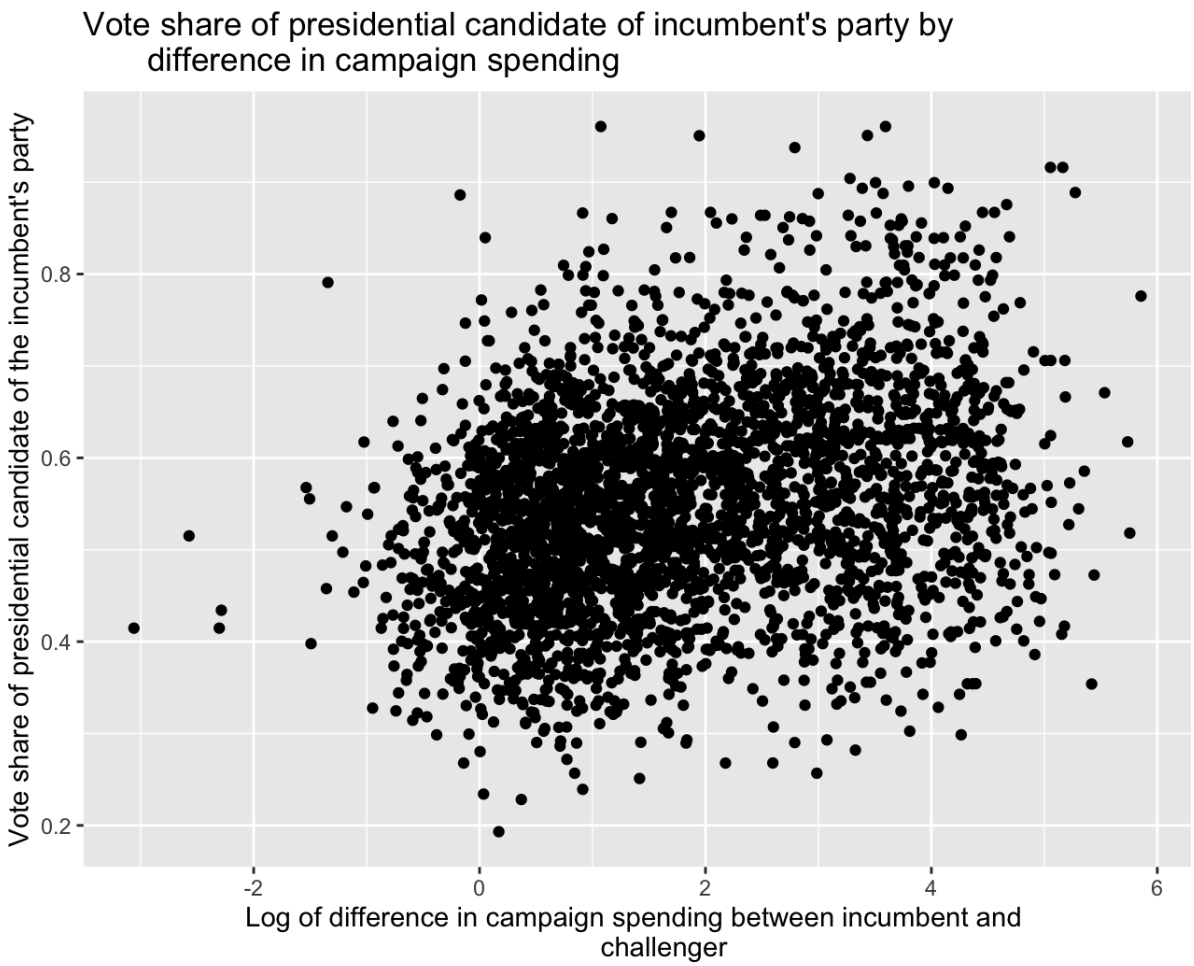
Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

   First, I create the scatterplot using the following code:

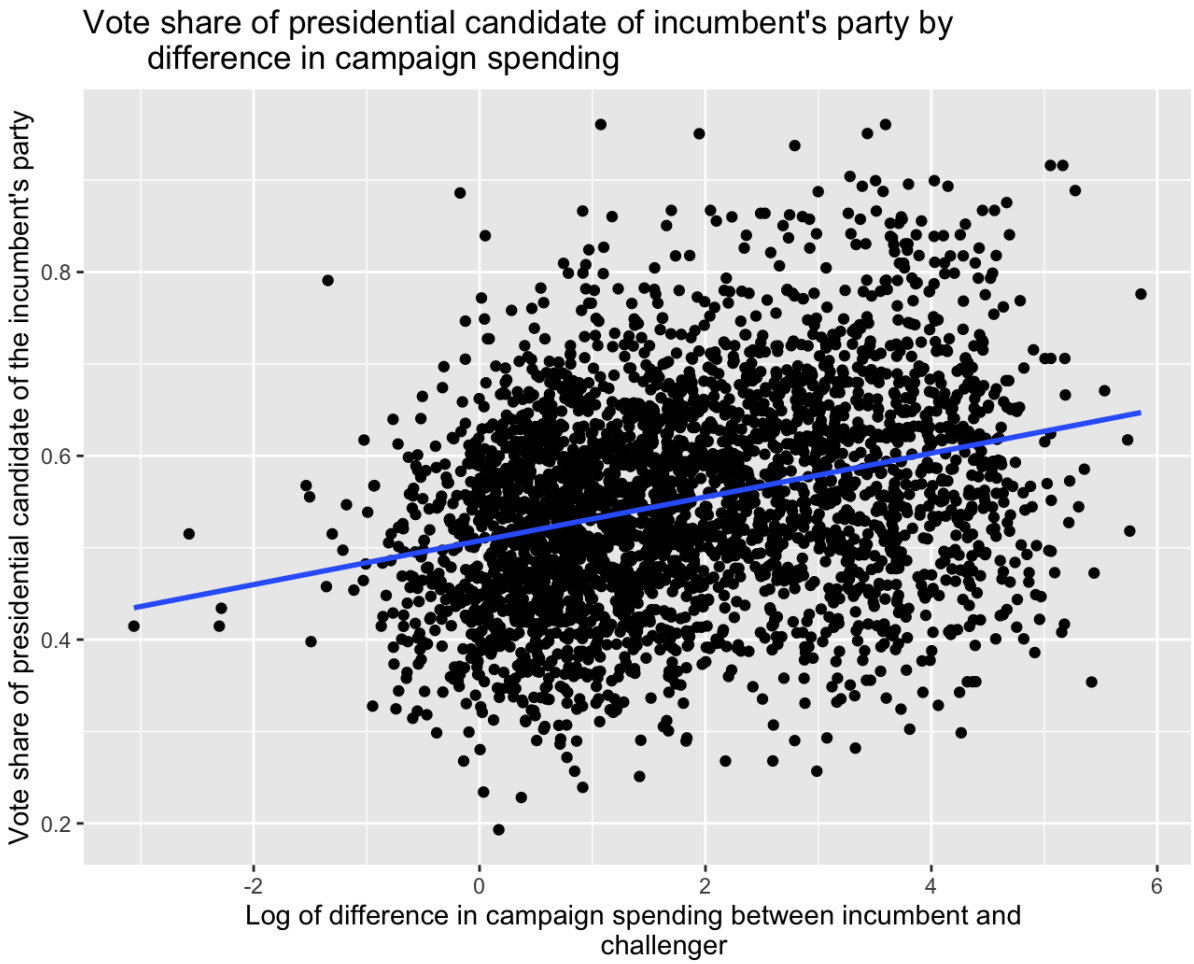
```
library(ggplot2)
ggplot(inc.sub, aes(x=difflog, y=presvote)) +
  geom_point() +
  labs(title = "Vote share of presidential candidate of incumbent's party by
      difference in campaign spending",
      x = "Log of difference in campaign spending between incumbent and
      challenger",
```

```
        y = "Vote share of presidential candidate of the incumbent's party
    ")
```



I add the regression line using this code:

```
ggplot(inc.sub, aes(x=difflog, y=presvote)) +
  geom_point() +
  labs(title = "Vote share of presidential candidate of incumbent's party
     by
        difference in campaign spending",
        x = "Log of difference in campaign spending between incumbent and
        challenger",
        y = "Vote share of presidential candidate of the incumbent's party
    ") +
  geom_smooth(
    aes(x=difflog, y=presvote),
    method = "lm",
    formula = y~x,
    se=FALSE)
```



3. Save the residuals of the model in a separate object.

   I save the residuals of the model in a separate object using this command:

```
residuals_presvote_difflog <- resid(reg_presvote_difflog)
```

4. Write the prediction equation.

   I again run the "summary" command to get the regression coefficient values:

```
summary(reg_presvote_difflog)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.32196 -0.07407 -0.00102  0.07151  0.42743

Coefficients:
```

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.507583   0.003161  160.60   <2e-16 ***
inc.sub$difflog 0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) = 0.507583

$\hat{\beta}_1$ (slope) = 0.023837

Therefore:

$\hat{presvote} = \mathbf{0.579031 + 0.041666(\textit{difflog})} + \epsilon$

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

   I regress *voteshare* on *presvote* using the following code:

```
reg_voteshare_presvote <- lm(inc.sub$voteshare ~ inc.sub$presvote, data=
    inc.sub)
reg_voteshare_presvote
summary(reg_voteshare_presvote)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.27330 -0.05888  0.00394  0.06148  0.41365

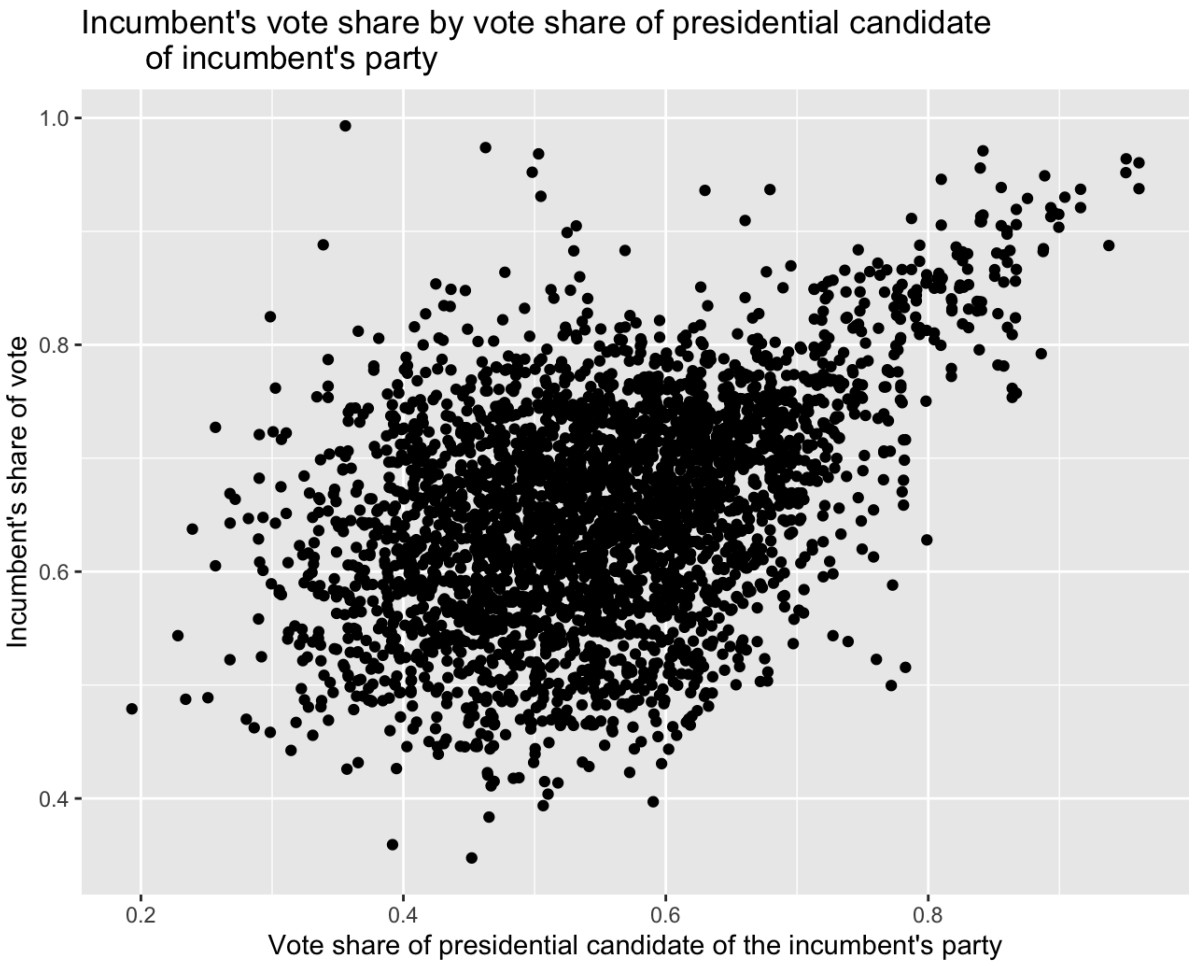
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.441330   0.007599   58.08   <2e-16 ***
inc.sub$presvote 0.388018   0.013493   28.76   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08815 on 3191 degrees of freedom
Multiple R-squared:  0.2058, Adjusted R-squared:  0.2056
F-statistic:   827 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

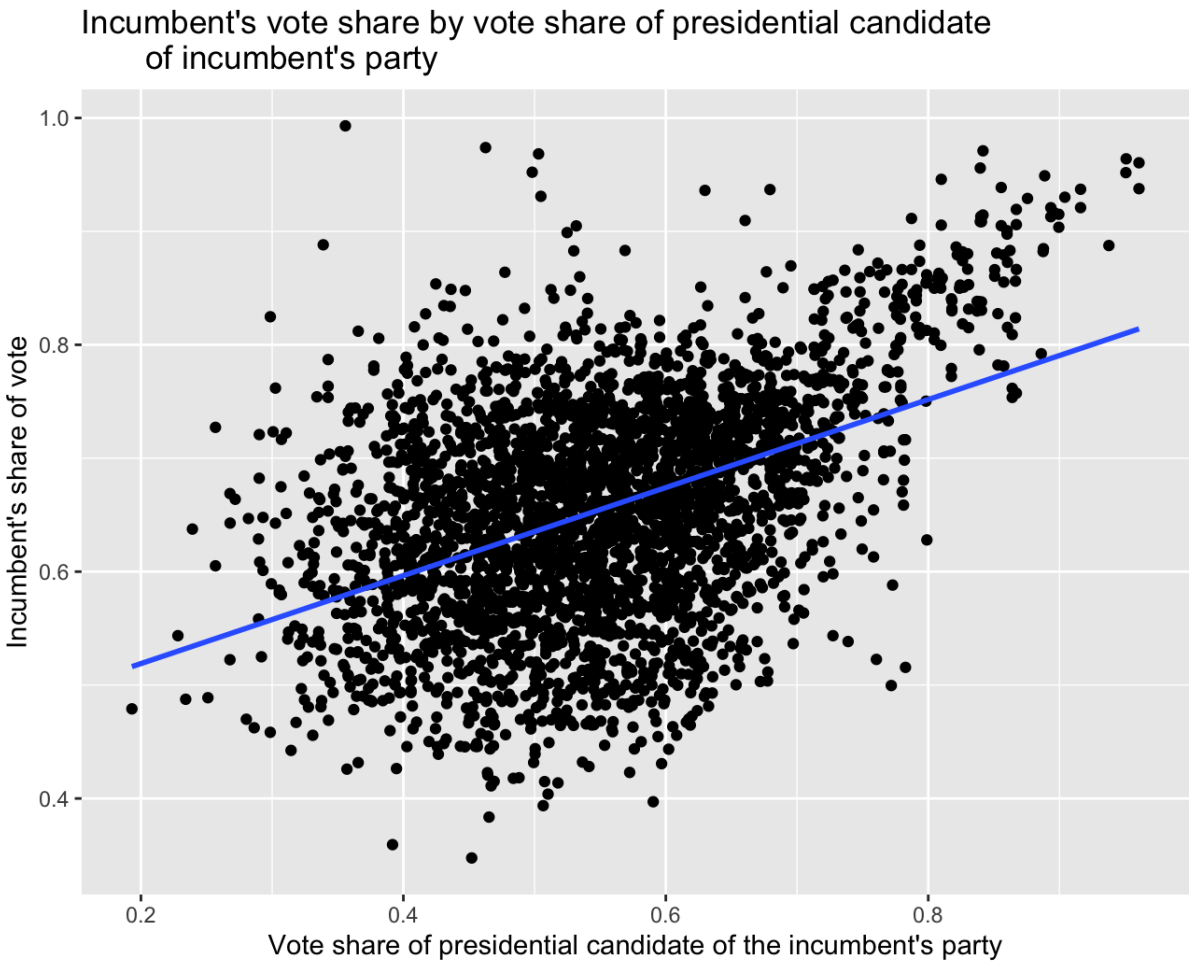
   First, I generate the scatterplot using the following code:

```
library(ggplot2)
ggplot(inc.sub, aes(x=presvote, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent's vote share by vote share of presidential
    candidate
       of incumbent's party",
       x = "Vote share of presidential candidate of the incumbent's party
    ",
       y = "Incumbent's share of vote")
```



I add the regression line using the following code:

```
ggplot(inc.sub, aes(x=presvote, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent's vote share by vote share of presidential
   candidate
       of incumbent's party",
       x = "Vote share of presidential candidate of the incumbent's party
   ",
       y = "Incumbent's share of vote") +
  geom_smooth(
    aes(x=presvote, y=voteshare),
    method = "lm",
    formula = y~x,
    se=FALSE)
```



3. Write the prediction equation.

   I again run the "summary" command to get the regression coefficient values:

```
summary(reg_voteshare_presvote)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.27330 -0.05888  0.00394  0.06148  0.41365

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.441330   0.007599   58.08   <2e-16 ***
inc.sub$presvote 0.388018   0.013493   28.76   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 0.08815 on 3191 degrees of freedom
Multiple R-squared:  0.2058, Adjusted R-squared:  0.2056
F-statistic:   827 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) = 0.441330

$\hat{\beta}_1$ (slope) = 0.388018

Therefore:

$$\hat{voteshare} = \mathbf{0.441330 + 0.388018(\boldsymbol{presvote})} + \epsilon$$

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

   I regress (the residuals from regressing *voteshare* on *difflog*) on (the residuals from regressing *presvote* on *difflog*) using the following code:

```
reg_residuals <- lm(residuals_voteshare_difflog ~ residuals_presvote_
    difflog)
reg_residuals
summary(reg_residuals)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)                -1.942e-18  1.299e-03    0.00        1
residuals_presvote_difflog  2.569e-01  1.176e-02   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

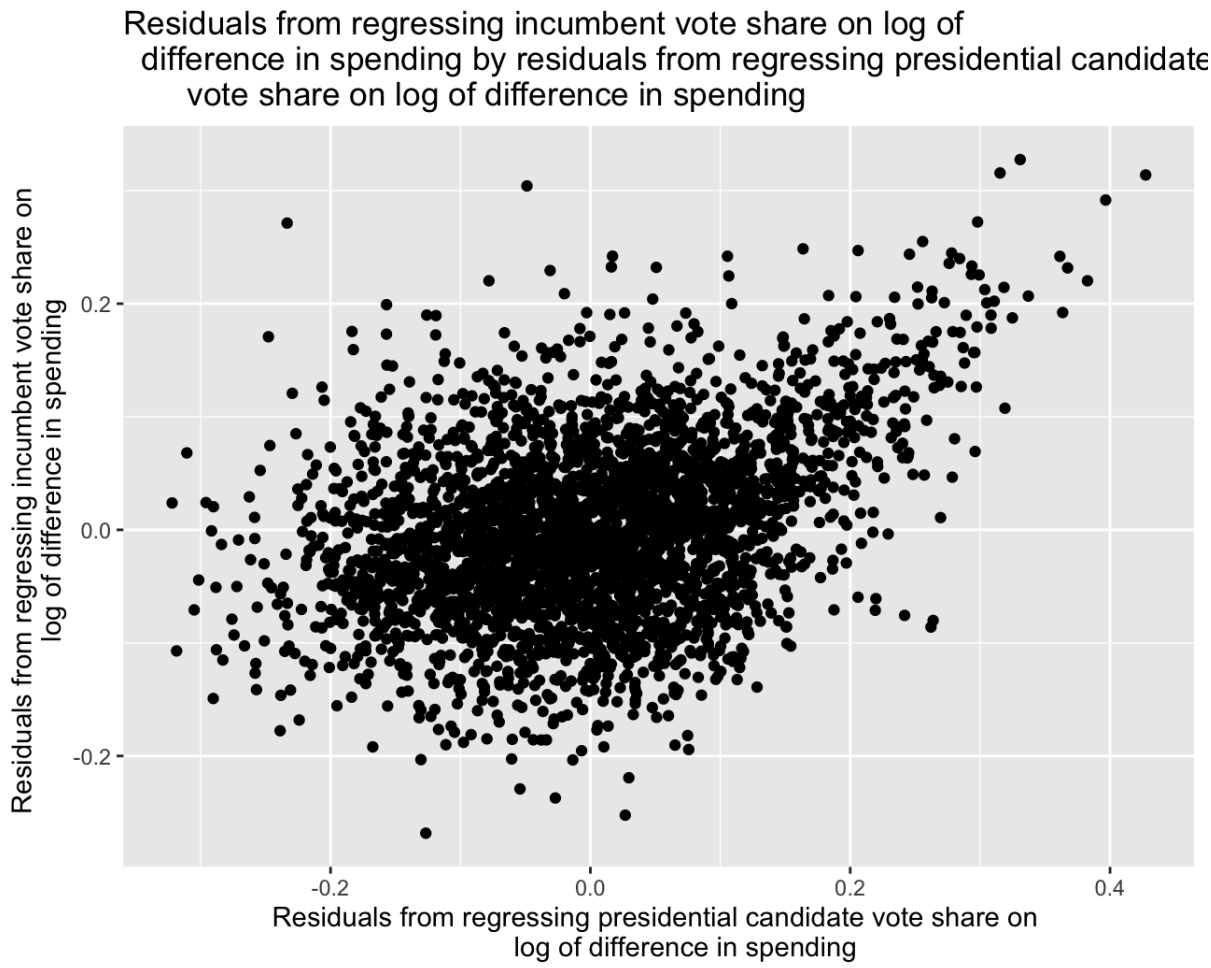
Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
F-statistic:   477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two residuals and add the regression line.

   First, I generate a scatterplot using the following code:

```
library(ggplot2)
ggplot(inc.sub, aes(x=residuals_presvote_difflog, y=residuals_voteshare_
    difflog)) +
  geom_point() +
  labs(title = "Residuals from regressing incumbent vote share on log of
```

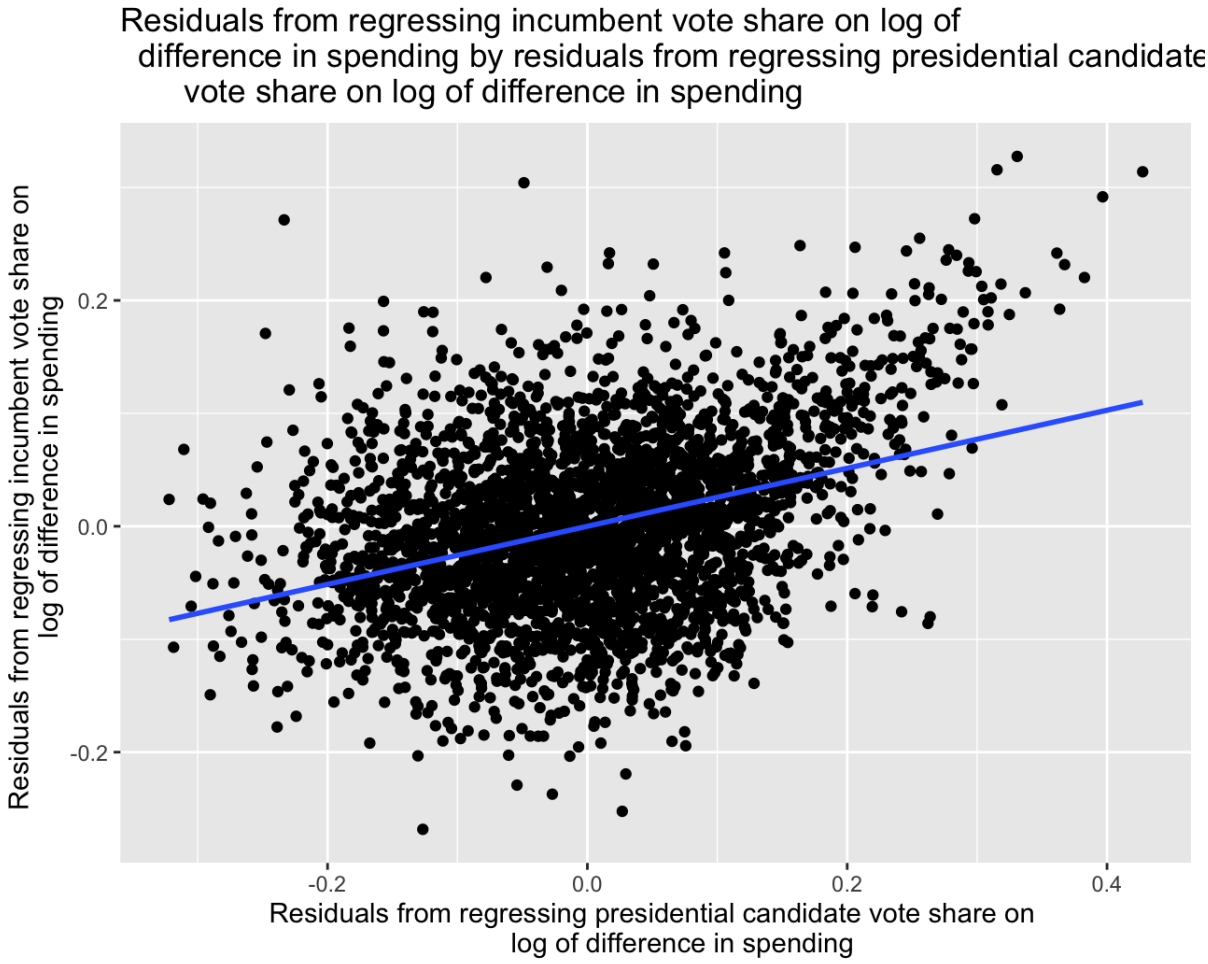
```
  difference in spending by residuals from regressing presidential
    candidate
        vote share on log of difference in spending",
        x = "Residuals from regressing presidential candidate vote share
    on
        log of difference in spending",
        y = "Residuals from regressing incumbent vote share on
        log of difference in spending")
```



I add the regression line using the following code:

```
ggplot(inc.sub, aes(x=residuals_presvote_difflog, y=residuals_voteshare_
    difflog)) +
  geom_point() +
  labs(title = "Residuals from regressing incumbent vote share on log of
  difference in spending by residuals from regressing presidential
    candidate
        vote share on log of difference in spending",
        x = "Residuals from regressing presidential candidate vote share
    on
        log of difference in spending",
```

```
        y = "Residuals from regressing incumbent vote share on
        log of difference in spending") +
  geom_smooth(
    aes(x=residuals_presvote_difflog, y=residuals_voteshare_difflog),
    method = "lm",
    formula = y~x,
    se=FALSE)
```



3. Write the prediction equation.

   I run the "summary" command again to get the regression coefficient values:

```
summary(reg_residuals)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126
```

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)                -1.942e-18  1.299e-03    0.00        1
residuals_presvote_difflog  2.569e-01  1.176e-02   21.84   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
F-statistic:   477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) = -1.942e-18

$\hat{\beta}_1$ (slope) = 2.569e-01

Therefore:

$\hat{residual from regressing vote share on difflog} = \mathbf{(-1.942e\text{-}18)} + \mathbf{(2.569e\text{-}01)}(\boldsymbol{residual\ from\ regressing\ presvote\ on\ difflog}) + \epsilon$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

   I regress *voteshare* on *difflog* and *presvote* using the following commands:

```
reg_voteshare_difflog_presvote <- lm(inc.sub$voteshare ~ inc.sub$difflog
    + inc.sub$presvote, data = inc.sub)
reg_voteshare_difflog_presvote
summary(reg_voteshare_difflog_presvote)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.4486442  0.0063297   70.88   <2e-16 ***
inc.sub$difflog  0.0355431  0.0009455   37.59   <2e-16 ***
inc.sub$presvote 0.2568770  0.0117637   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07339 on 3190 degrees of freedom
Multiple R-squared:  0.4496, Adjusted R-squared:  0.4493
F-statistic:  1303 on 2 and 3190 DF,  p-value: < 2.2e-16
```

2. Write the prediction equation.

   From the output above:

   $\hat{\beta}_0 = 0.4486442$

   $\hat{\beta}_1 = 0.0355431$

   $\hat{\beta}_2 = 0.2568770$

   Therefore:

   $$\hat{voteshare} = \mathbf{0.4486442 + 0.0355431(\textit{difflog}) + 0.2568770(\textit{presvote})} + \epsilon$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

   From the output tables above, I see that the minimum, first quartile, median, third quartile, and maximum residuals are identical:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126
```

   The residual standard errors are also identical:

```
Residual standard error: 0.07339
```

   ??????????????OK BUT WHY ARE THEY IDENTICAL????????????????????